NHANES Hypertension Project: Visualizations

Figures supporting the project-defined measured high-blood-pressure classification. This outcome is not a clinical diagnosis. Exploratory plots and predictive evaluation are unweighted sample-based unless stated otherwise.

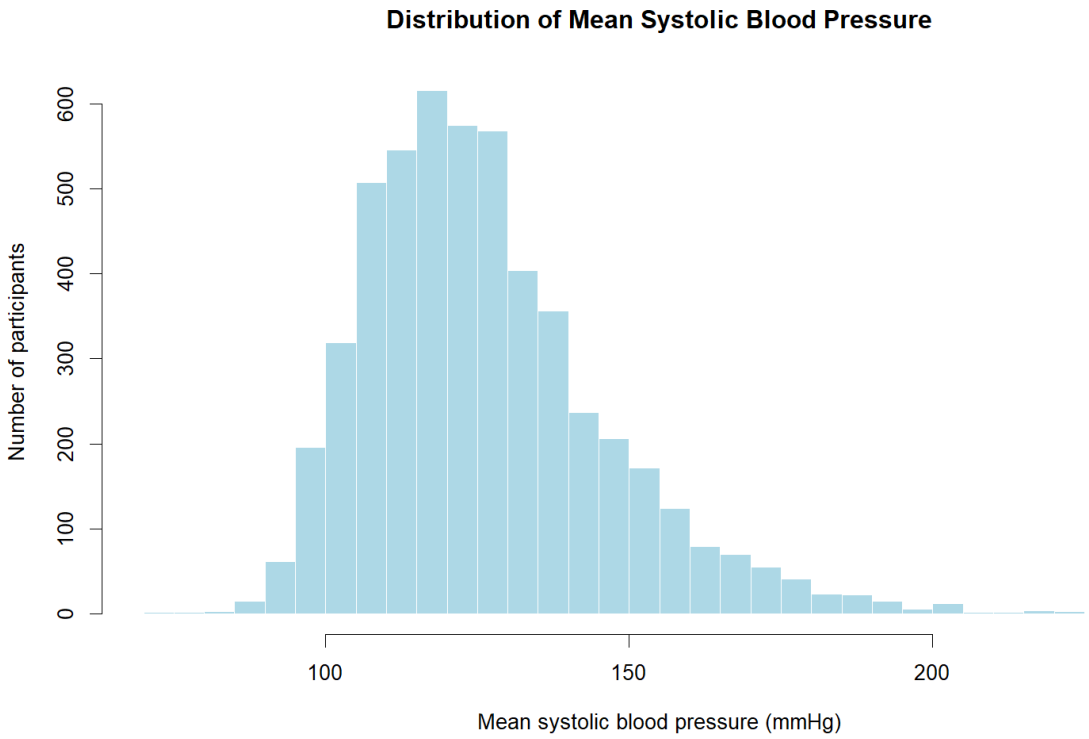


Figure 1. Distribution of mean systolic blood pressure

The distribution is concentrated around the lower-to-middle range, with a right tail showing fewer participants at higher mean systolic readings.

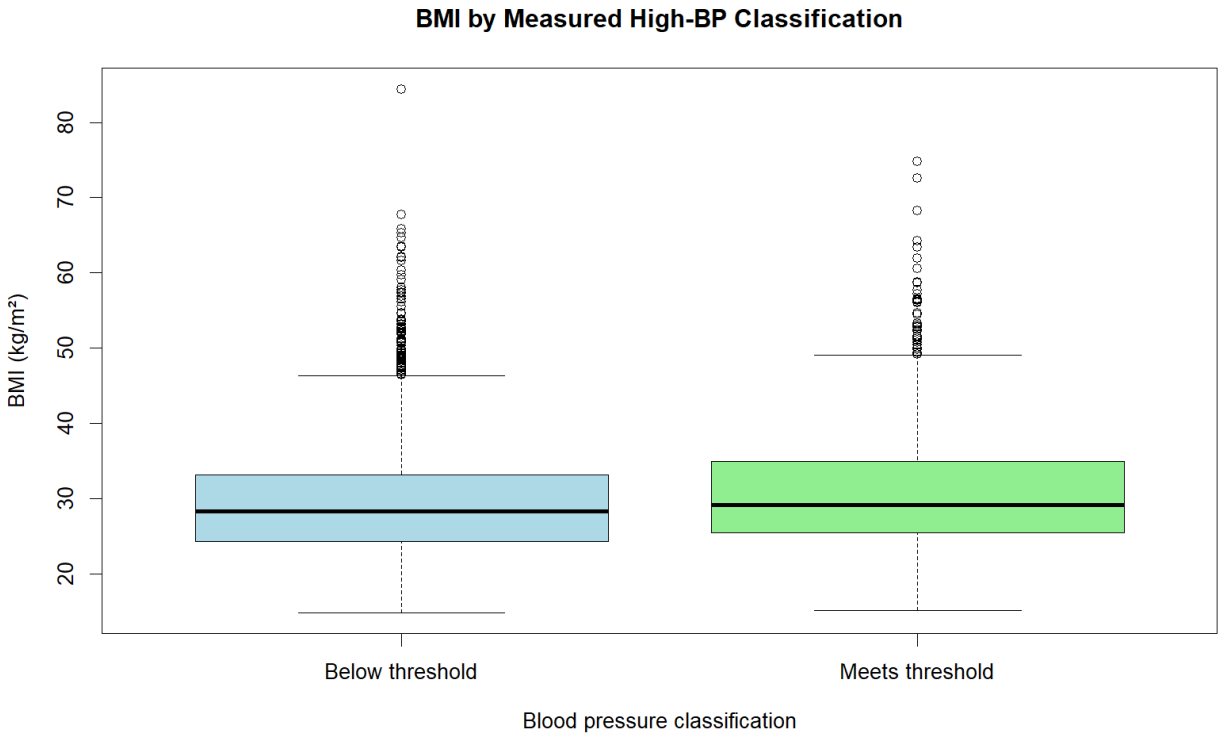


Figure 2. BMI by project-defined measured high-BP classification

The outcome-positive group has a somewhat higher median BMI. The distributions overlap substantially, so BMI alone does not distinguish the groups.

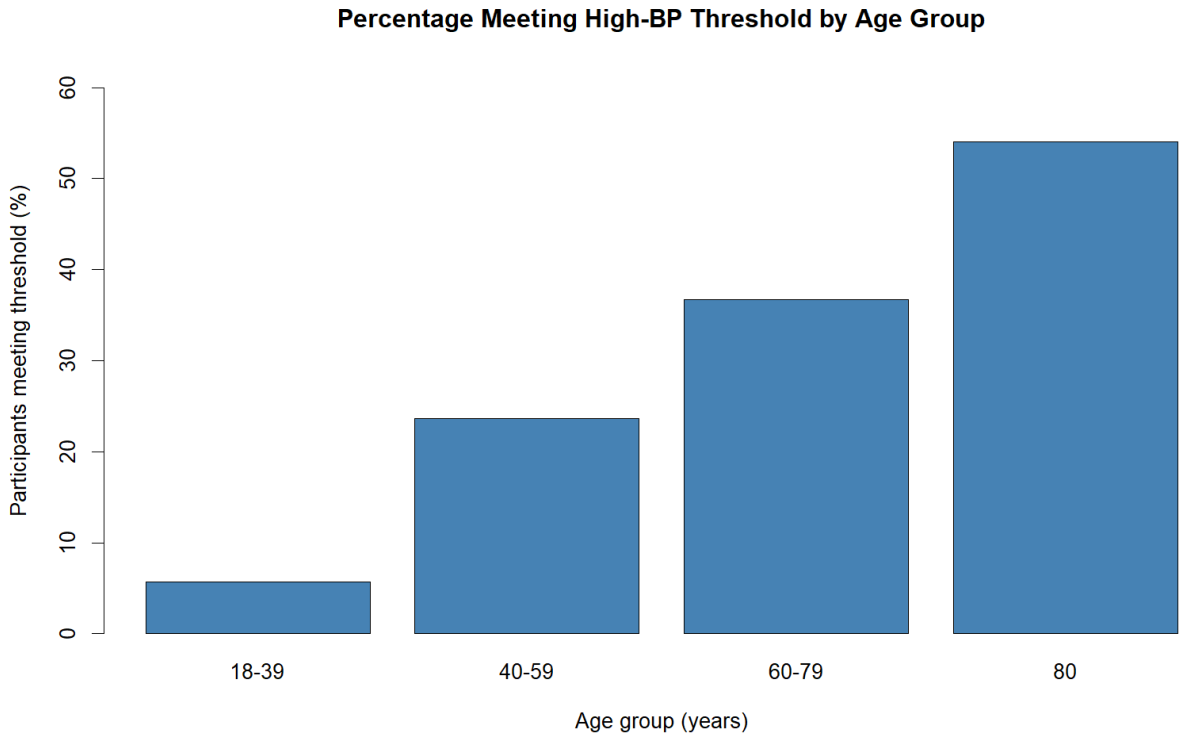


Figure 3. Percentage meeting the project-defined high-BP threshold by age group

The observed percentage increases across age groups in this unweighted sample. The age-80 group has a smaller sample and missing outcomes should be kept in mind.

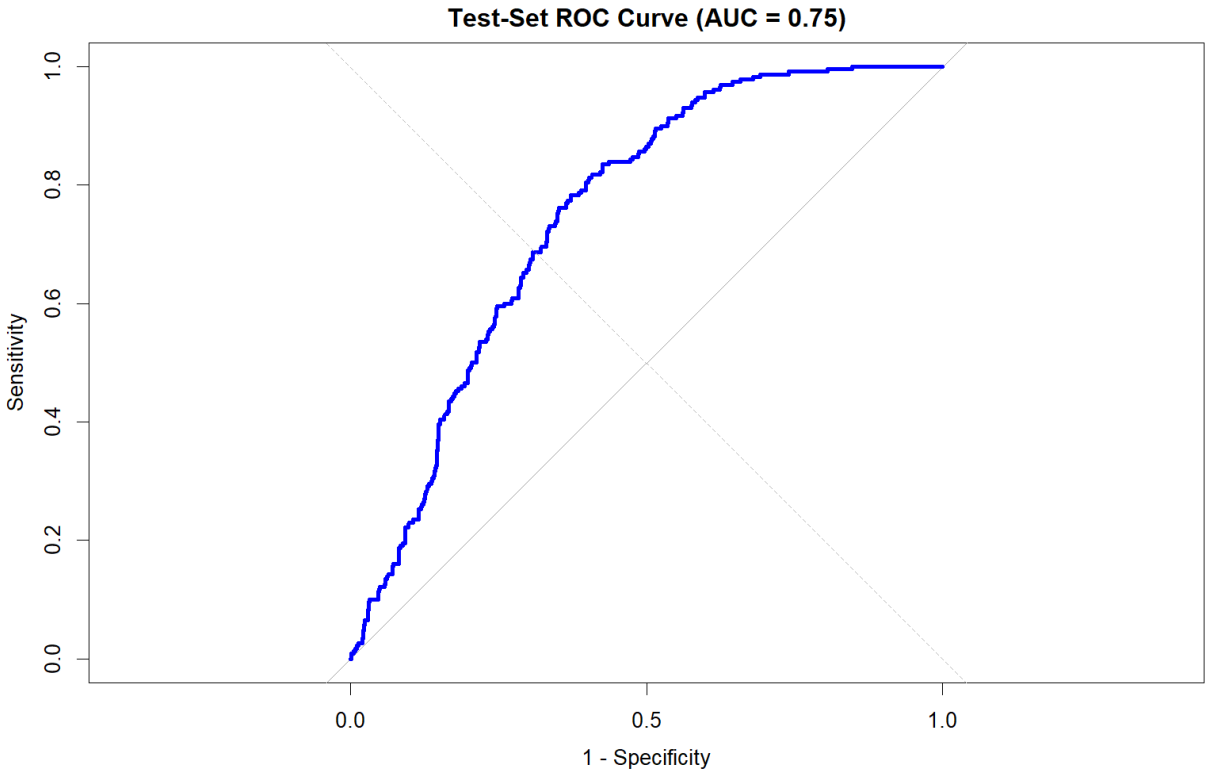


Figure 4. ROC curve for the held-out test set

The test-set AUC is 0.75, indicating moderate discrimination in this sample. These predictive metrics are unweighted.

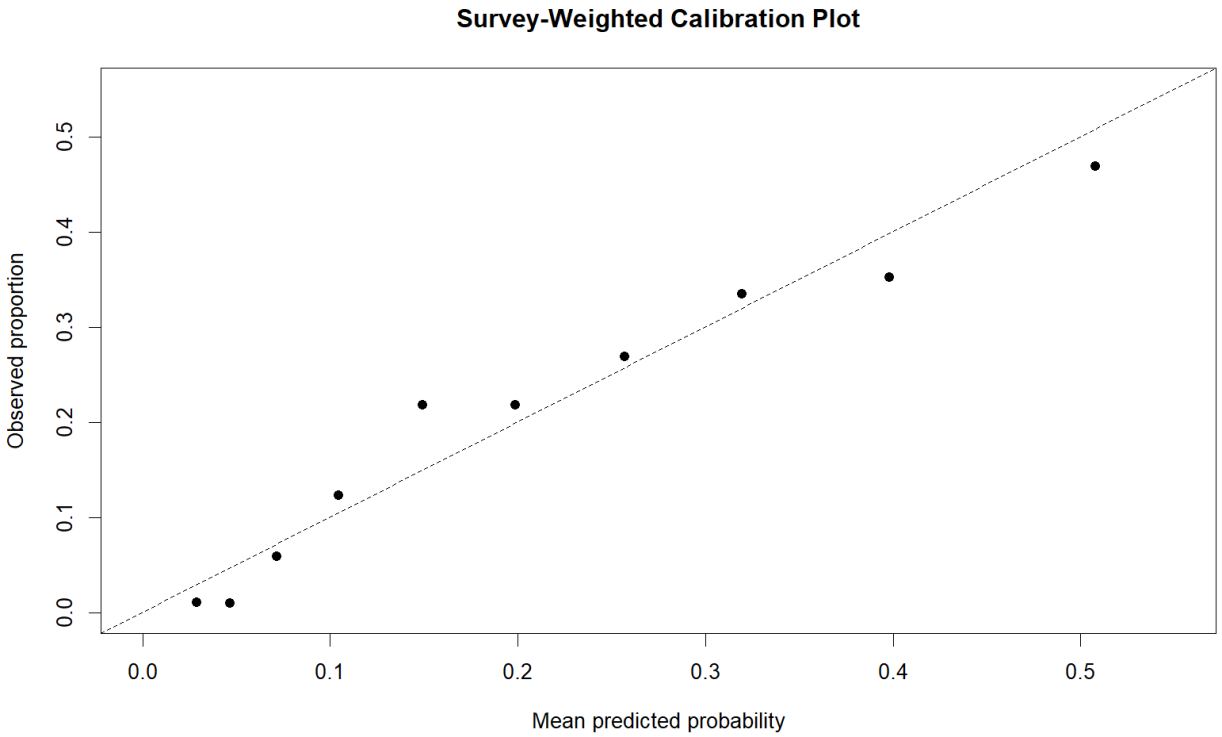


Figure 5. Calibration plot

Points nearer the diagonal indicate closer agreement between predicted and observed proportions. The plot should be interpreted in light of the calibration grouping and weighting procedure used.